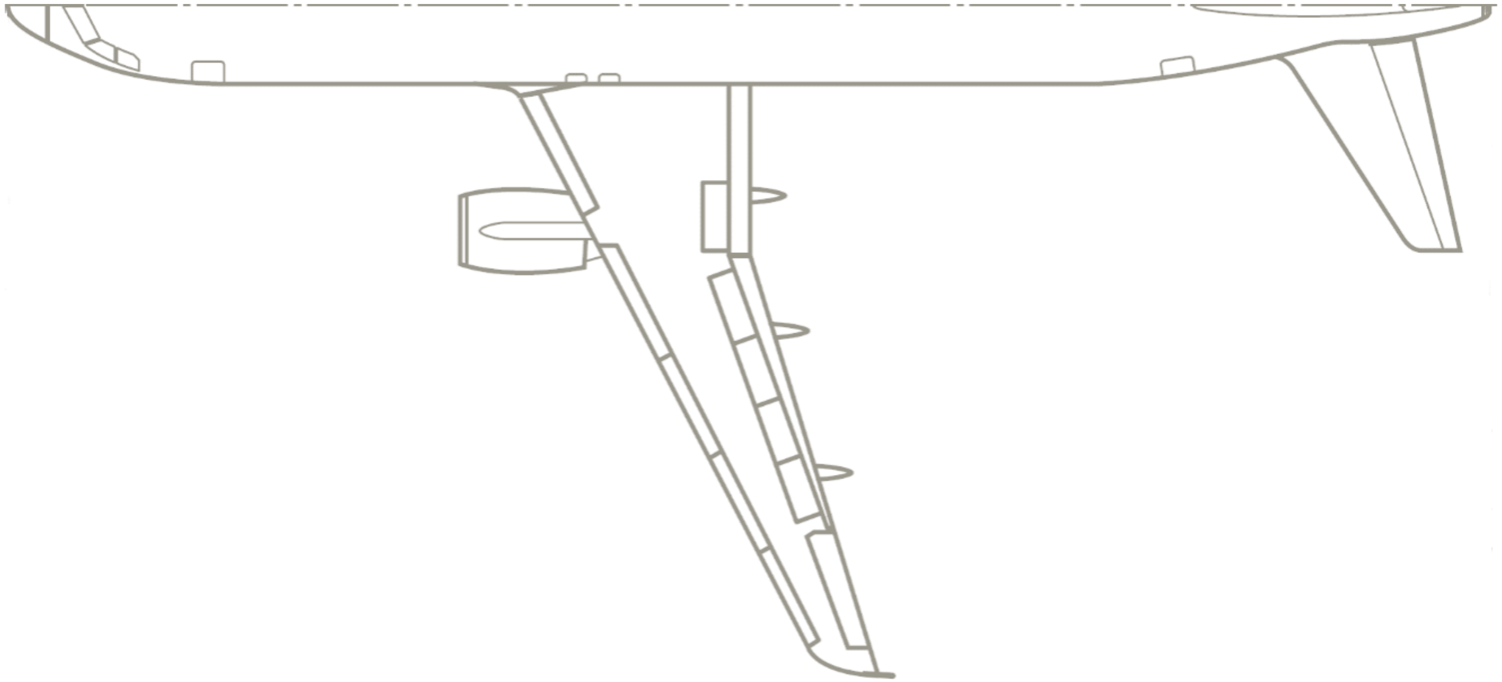
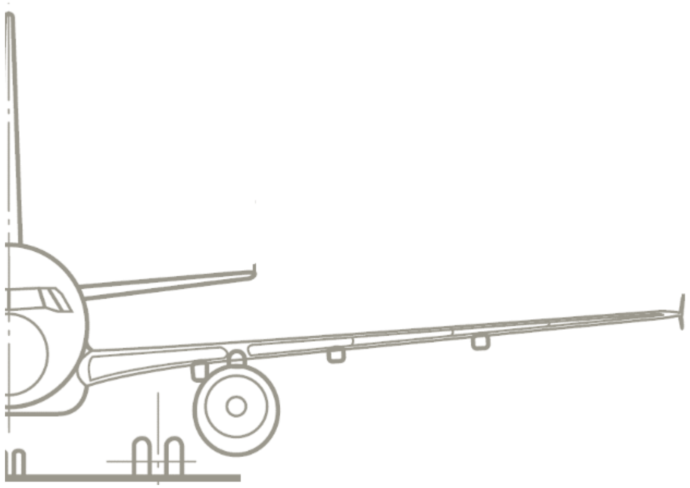
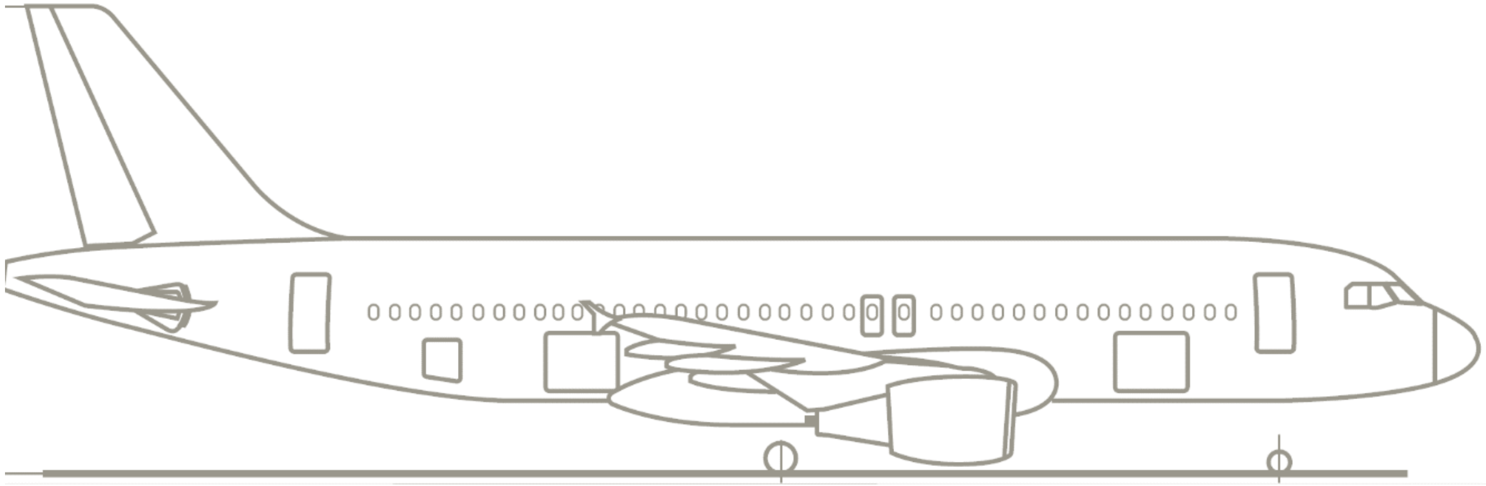


Aircraft Design

Tutorial 1 - Lift





Tutorial Schedule

Dates	Planned tutorials	Tutorial topic
28.10.2022	Tutorial 1	Lift
04.11.2022	Tutorial 2	Drag
25.11.2022	Tutorial 3	Stability
02.12.2022	Tutorial 4	Mass estimation
16.12.2022	Tutorial 5	Payload-Range Diagram
23.12.2022	Tutorial 6	Take-off and landing
Holiday break		
13.01.2023	Tutorial 7	Design Chart
20.01.2023	Tutorial 8	Configurations & Design Process
03.02.2023	Exam preparation	Revision
Exam **14.02.2023 at 11:30 am**		

Chapter 2

Chapter 3

Chapters 4, 6

Chapter 7

All chapters

➢ Time schedule also posted in Moodle!

Tutorial Format

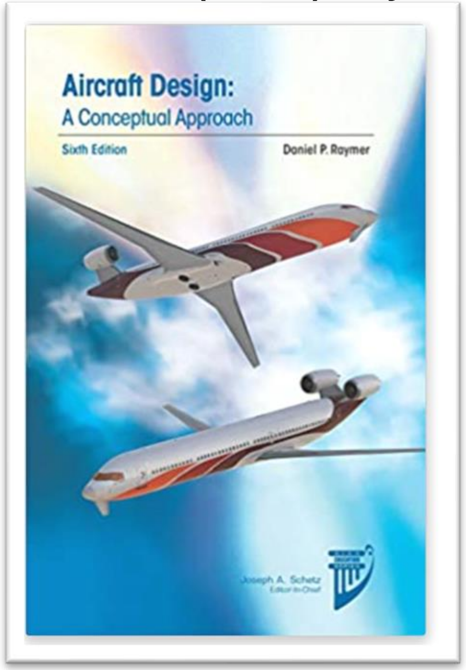
- Theory questions & calculation tasks
  - ➢ All tutorials (except the last one) will be calculated using the A320 dataset provided in *Moodle;* use it whenever values are given
- Solutions for calculation tasks will be provided on Moodle; solutions for theory questions not!

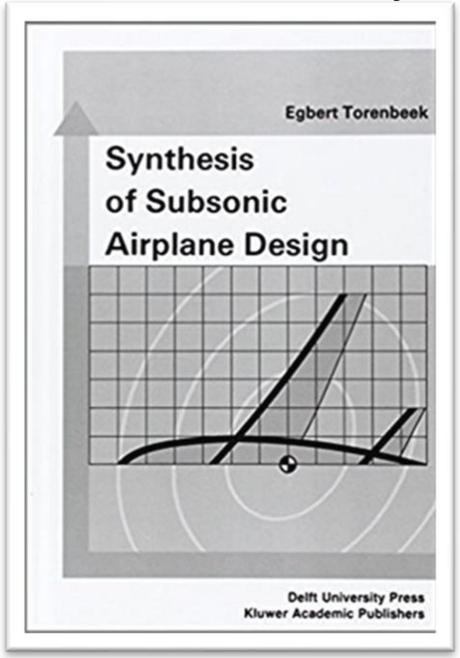
- ***MATLAB live scipts*** of the tutorials will give you the opportunity to unterstand correlations better (uploaded after the tutorials)
- In every tutorial we will have a short break of 5min
  - ➢ Please post all questions in the forum on *Moodle* such that everyone has access to the answers!

General Information

Literature

- Raymer, D.P. (2018). Aircraft design: a conceptual approach. Reston, VA: AIAA Education.
- Roskam, J. (2003). Airplane design Vol. I-VIII. Ottawa, KS: DARcorporation.
- Torenbeek, E. (1982). Synthesis of subsonic airplane design. Delft, NLD: Delft University Press.



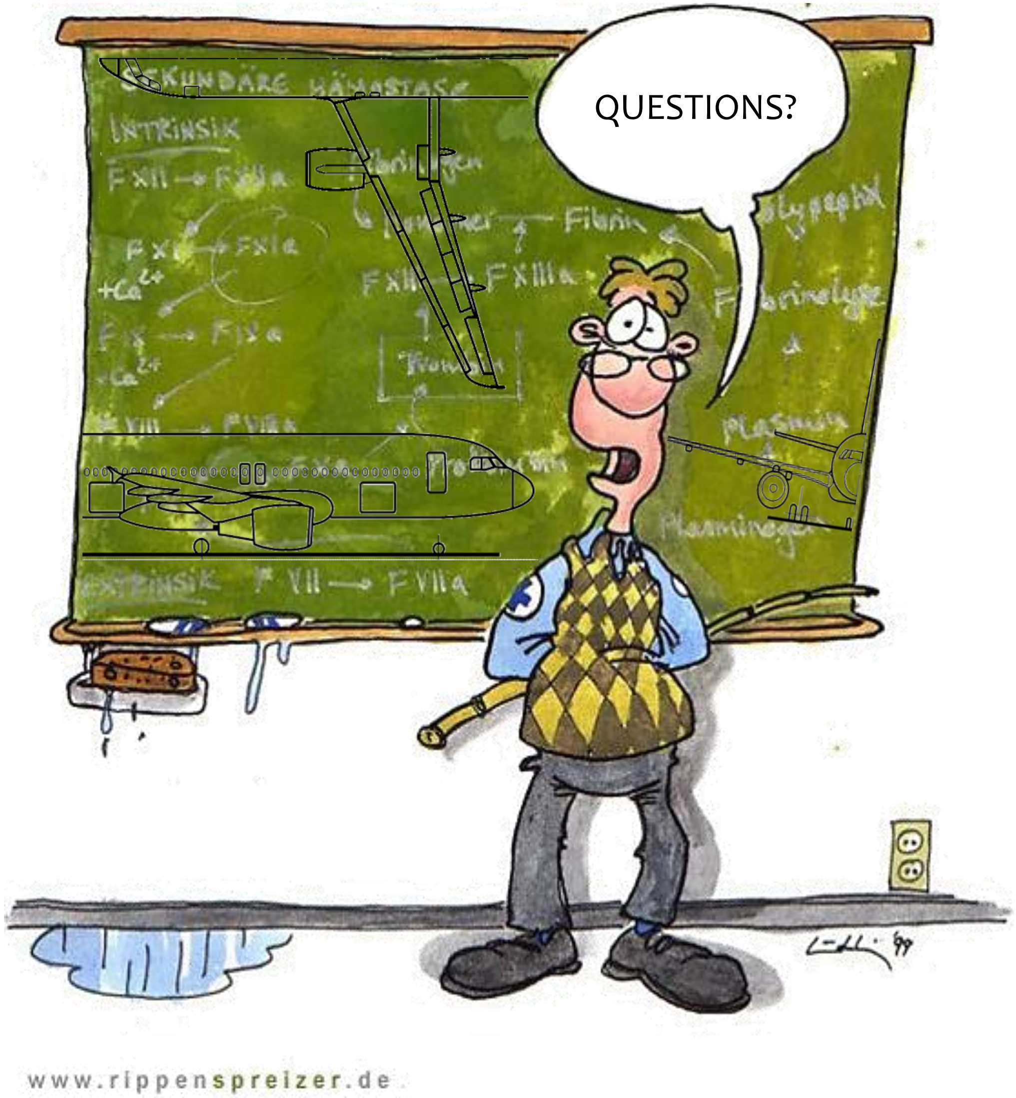


➢ Background to used methods in tutorial (for further studies)

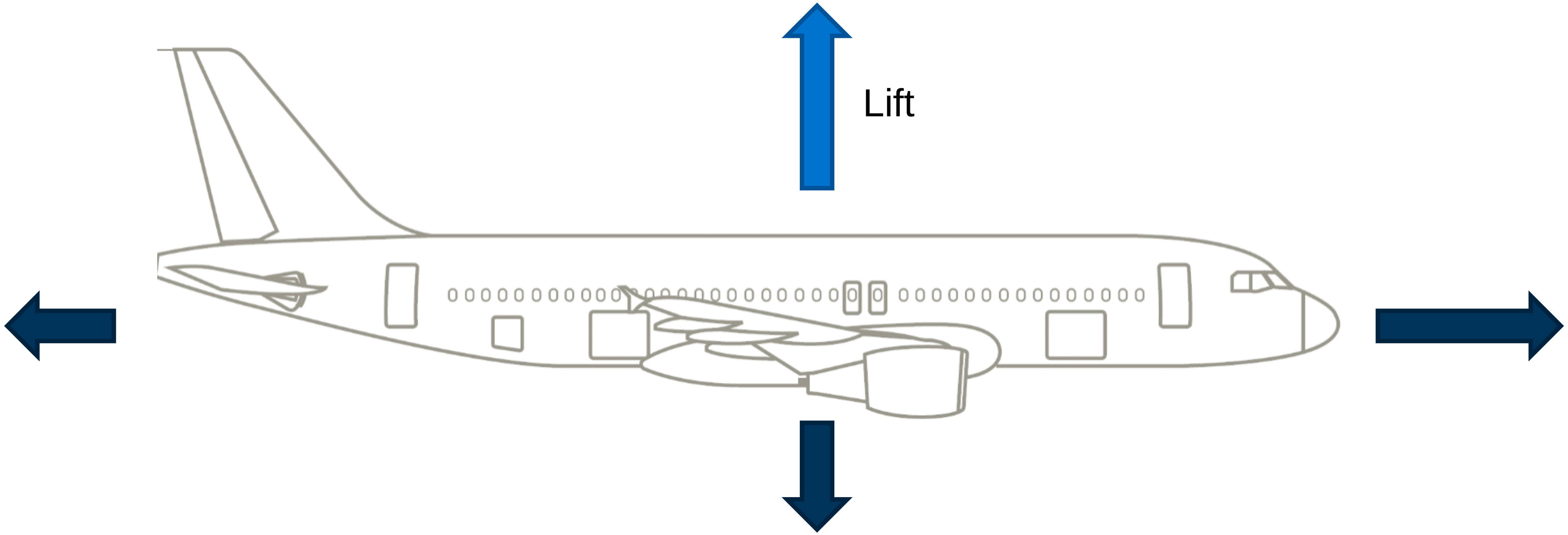
General Information

Exam

- Duration: 90 min
- Style: - Written exam
  - Mixture of questions (of style multiple-choice & „free-text“) and calculation tasks (~ 50/50)
- Aids: - Except from a pen, ruler and calculator (not programmable) no other aids are allowed!
  - A dictionary may be used.
- Other hints: - List with **equations to be known by heart** on Moodle
  - Unit conversion factors will be given!
  - You can answer in English or in German
  - Please give numeric values with **four meaningful** decimal places (0.1234, 1.234, 123400)


QUESTIONS?
www.rippenspreizer.de

Tutorial 1 - Lift



Basics

- 🕮 *Why is calculation of lift characteristics important in aircraft design?*

- 🕮 *What methods can be applied in order to set up an aircraft lift curve?*

- 🕮 *What are the basic configurational & operational factors influencing aircraft lift?*

Task 1

Determine the aspect ratio AR and taper ratio λ of the configuration with the given values in the data set. Compare the results with those given by the manufacturer and explain the differences.

<u>Cause for the difference:</u>

Task 2

Estimate the lift coefficient slope $C_{L\alpha}$ *of the finite wings according to the lifting line theory.*

Task 3

For a more accurate result, $C_{L\alpha}$ is to be calculated by the equation of Polhamus at the cruise Mach number of $M = 0.78$. Name a possible cause for the difference to the previously calculated value.

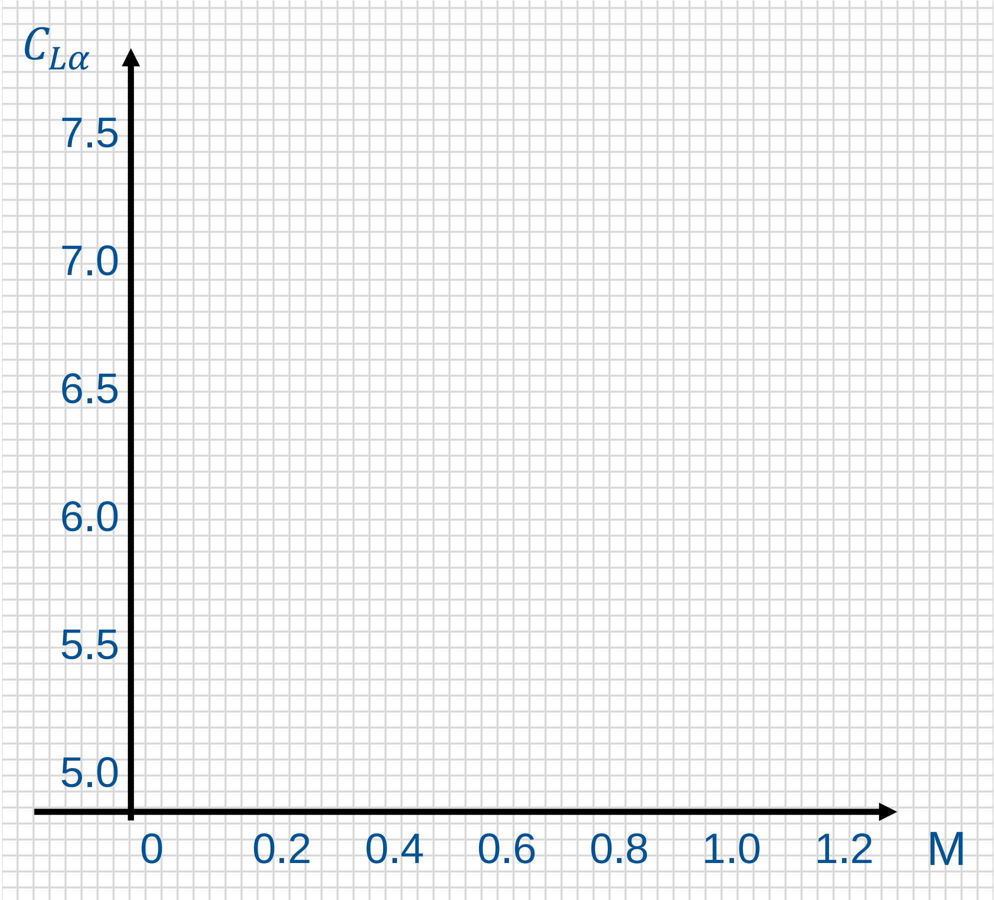
Task 4

Plot $C_{L\alpha}$ against the Mach number for the region $M = 0.2 \ldots 0.8$ with the help of one calculation method (choose the most appropriate one). Extend the plot for the transonic regime ($M = 0.8 \ldots 0.95$) and explain the curve path.

M [−]	$C_{L\alpha}$ [rad^{-1}]
0.2	
0.3	
0.4	
0.5	
0.6	
0.7	
0.8	

Task 4

Plot $C_{L\alpha}$ against the Mach number for the region $M = 0.2 \ldots 0.8$ with the help of one calculation method (choose the most appropriate one). Extend the plot for the transonic regime ($M = 0.8 \ldots 0.95$) and explain the curve path.



<u>Explanation for the transonic regime curve:</u>

Task 5

Plot the lift curve (C_L vs. α) for low Mach number (use the estimation of $C_{L\alpha}$ from Task 2 and given geometrical data in the data set). Assume an aircraft stall angle of $\alpha_{max} = 10°$.

Hint → In order to plot the lift curve, three parameters are necessary:

Task 5

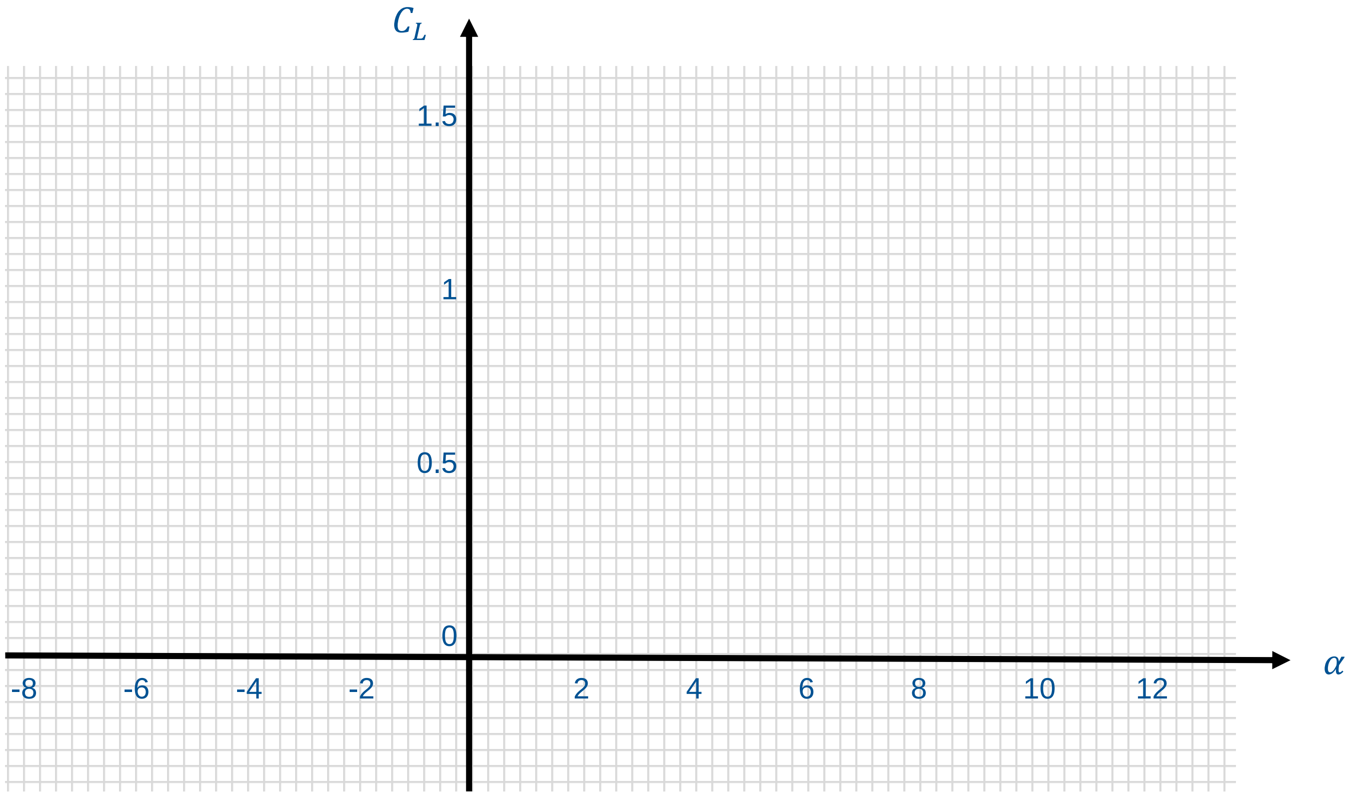
Plot the lift curve (C_L vs. α) for low Mach number (use the estimation of $C_{L\alpha}$ from Task 2 and given geometrical data in the data set). Assume an aircraft stall angle of $\alpha_{max} = 10°$.

Task 5

Plot the lift curve (C_L vs. α) for low Mach number (use the estimation of $C_{L\alpha}$ from Task 2 and given geometrical data in the data set). Assume an aircraft stall angle of $\alpha_{max} = 10°$.

Task 5

Plot the lift curve (C_L vs. α) by using the estimation of $C_{L\alpha}$ from task 2 and given geometrical data in the data set. Assume an aircraft stall angle of $\alpha_{max} = 10°$.



Break (5min)

This video is not relevant for the exam!

Wing-in-ground-effect aircraft

Source: https://www.youtube.com/watch?v=QMzGtUMv5j0

Task 6

Determine the reference 1g stalling speed V_{S1g} at maximum take-off weight in clean configuration during level flight at sea level conditions. Draw the free body diagram of the aircraft in this situation.

Free-body diagram:

Task 7

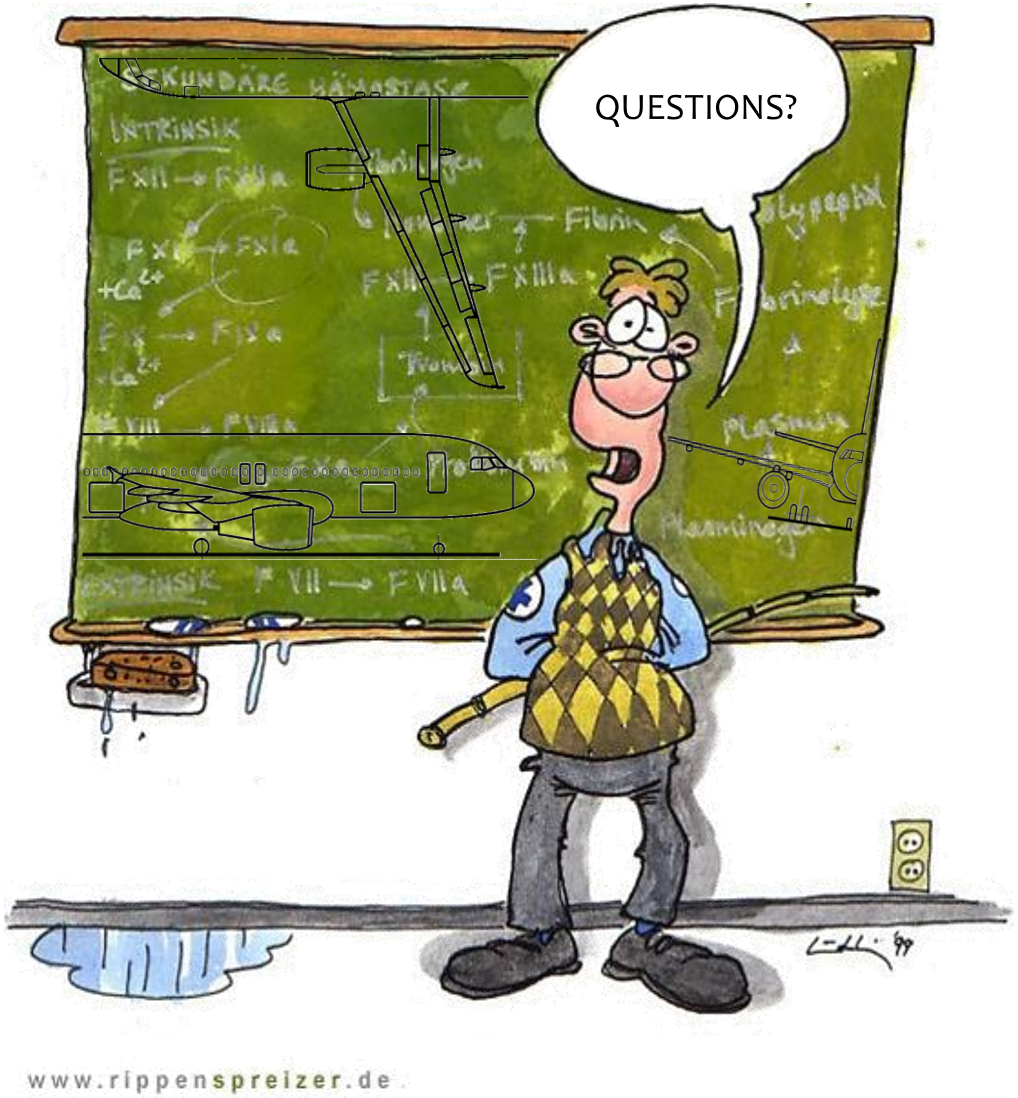
Determine the aircraft angle of attack α_C *at a cruise speed with* $M_C = 0.78$ *at 90% of the maximum take-off weight at an altitude of* $h = 12km$. *Assume* $\alpha_{max} = 6°$.

Task 7

Determine the aircraft angle of attack α_C *at a cruise speed with* $M_C = 0.78$ *at 90% of the maximum take-off weight at an altitude of* $h = 12km$. *Assume* $\alpha_{max} = 6°$.

Task 7

Determine the aircraft angle of attack α_C *at a cruise speed with* $M_C = 0.78$ *at 90% of the maximum take-off weight at an altitude of* $h = 12km$. *Assume* $\alpha_{max} = 6°$.


QUESTIONS?
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